

1. In this quiz, you will calculate the Hessian for some functions of 2 variables and functions of 3 variables.

1 point

For the function $f(x, y) = x^3y + x + 2y$, calculate the Hessian matrix $H = \begin{bmatrix} \partial_{x,x}f & \partial_{x,y}f \\ \partial_{y,x}f & \partial_{y,y}f \end{bmatrix}$

- ☒ $H = \begin{bmatrix} 6xy & 3x^2 \\ 3x^2 & 0 \end{bmatrix}$
- ☐ $H = \begin{bmatrix} 6xy & -3x^2 \\ -3x^2 & 0 \end{bmatrix}$
- ☐ $H = \begin{bmatrix} 0 & -3x^2 \\ -3x^2 & 6xy \end{bmatrix}$
- ☐ $H = \begin{bmatrix} 0 & 3x^2 \\ 3x^2 & 6xy \end{bmatrix}$

2. For the function $f(x, y) = e^x \cos(y)$, calculate the Hessian matrix.

1 point

- ☒ $H = \begin{bmatrix} e^x \cos(y) & -e^x \sin(y) \\ -e^x \sin(y) & -e^x \cos(y) \end{bmatrix}$
- ☐ $H = \begin{bmatrix} -e^x \cos(y) & -e^x \sin(y) \\ e^x \sin(y) & -e^x \cos(y) \end{bmatrix}$
- ☐ $H = \begin{bmatrix} -e^x \cos(y) & -e^x \sin(y) \\ -e^x \sin(y) & e^x \cos(y) \end{bmatrix}$
- ☐ $H = \begin{bmatrix} -e^x \cos(y) & e^x \sin(y) \\ -e^x \sin(y) & -e^x \cos(y) \end{bmatrix}$

3. For the function $f(x, y) = \frac{x^2}{2} + xy + \frac{y^2}{2}$, calculate the Hessian matrix.

1 point

Notice something interesting when you calculate $\frac{1}{2}[x, y]H \begin{bmatrix} x \\ y \end{bmatrix}$!

- ☐ $H = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
- ☐ $H = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$
- ☐ $H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- ☒ $H = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

4. For the function $f(x, y, z) = x^2 e^{-y} \cos(z)$, calculate the Hessian matrix

1 point

$$H = \begin{bmatrix} \partial_{x,x}f & \partial_{x,y}f & \partial_{x,z}f \\ \partial_{y,x}f & \partial_{y,y}f & \partial_{y,z}f \\ \partial_{z,x}f & \partial_{z,y}f & \partial_{z,z}f \end{bmatrix}$$

- ☐ $H = \begin{bmatrix} 2e^{-y} \cos(z) & 2xe^{-y} \cos(z) & 2xe^{-y} \sin(z) \\ 2xe^{-y} \cos(z) & x^2 e^{-y} \cos(z) & x^2 e^{-y} \sin(z) \\ 2xe^{-y} \sin(z) & x^2 e^{-y} \sin(z) & x^2 e^{-y} \cos(z) \end{bmatrix}$
- ☒ $H = \begin{bmatrix} 2e^{-y} \cos(z) & -2xe^{-y} \cos(z) & -2xe^{-y} \sin(z) \\ -2xe^{-y} \cos(z) & x^2 e^{-y} \cos(z) & x^2 e^{-y} \sin(z) \\ -2xe^{-y} \sin(z) & x^2 e^{-y} \sin(z) & -x^2 e^{-y} \cos(z) \end{bmatrix}$
- ☐ $H = \begin{bmatrix} 2xe^{-y} \cos(z) & x^2 e^{-y} \cos(z) & 2xe^{-y} \sin(z) \\ 2xe^{-y} \cos(z) & x^2 e^{-y} \cos(z) & x^2 xe^{-y} \sin(z) \\ 2xe^{-y} \sin(z) & 2xe^{-y} \sin(z) & 2xe^{-y} \cos(z) \end{bmatrix}$
- ☐ $H = \begin{bmatrix} 2xe^{-y} \cos(z) & -2e^{-y} \cos(z) & -2e^{-y} \sin(z) \\ -2e^{-y} \cos(z) & x^2 e^{-y} \cos(z) & x^2 e^{-y} \sin(z) \\ -2x^2 e^{-y} \sin(z) & x^2 e^{-y} \sin(z) & -2xe^{-y} \cos(z) \end{bmatrix}$

5. For the function $f(x, y, z) = xe^y + y^2 \cos(z)$, calculate the Hessian matrix.

1 point

- ☐ $H = \begin{bmatrix} 0 & e^y & 0 \\ e^y & xe^y + 2\sin(z) & -2y \cos(z) \\ 0 & -2y \cos(z) & -y^2 \sin(z) \end{bmatrix}$
- ☐ $H = \begin{bmatrix} 0 & e^y & 0 \\ e^y & xe^y + 2\sin(z) & 2y \cos(z) \\ 0 & 2y \cos(z) & y^2 \sin(z) \end{bmatrix}$
- ☒ $H = \begin{bmatrix} 0 & e^y & 0 \\ e^y & xe^y + 2\cos(z) & -2y \sin(z) \\ 0 & -2y \sin(z) & -y^2 \cos(z) \end{bmatrix}$



$$H = \begin{bmatrix} 0 & e'' & 0 \\ e^y & xe^y + 2\cos(z) & 2ysin(z) \\ 0 & 2ysin(z) & y^2\cos(z) \end{bmatrix}$$